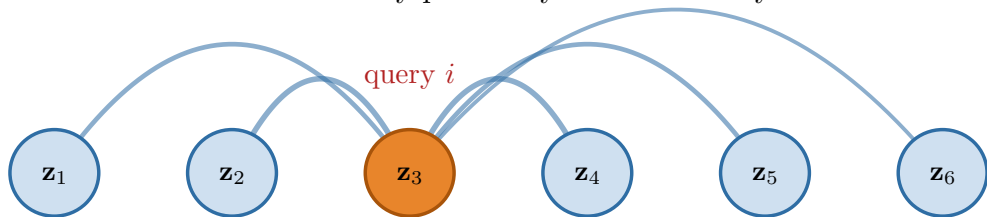


self-attention: every pulse may attend to every other



$$a_{ij} = \frac{\exp(\mathbf{q}_i \cdot \mathbf{k}_j / \sqrt{d_k})}{\sum_{\ell} \exp(\mathbf{q}_i \cdot \mathbf{k}_{\ell} / \sqrt{d_k})}$$

$$\mathbf{z}_i^{(1)} = \mathbf{z}_i^{(0)} + \sum_j a_{ij} \mathbf{v}_j$$